



Figure 2: Simple Logging Facade for Java



Figure 3: Stable JAR files from SLF4J portal

suggestive modelling is known as a recommender engine or recommendation system. Even on YouTube, we get a number of suggestions related to videos that we viewed earlier. Such online platforms integrate the approaches of recommendation engines, as a result of which the related best fit or most viewed items are presented to the user as recommendations.

Apache Mahout provides the platform to program and implement recommender systems. For example, the Twitter hashtag popularity can be evaluated and ranked based on the visitor count, popularity or simply the hits by the users. In YouTube, the number of viewers is the key value that determines the actual popularity of that particular video. Such algorithms can be implemented using Apache Mahout, which are covered under high performance real-time machine learning.

For example, a data table that presents the popularity of products after online shopping by consumers is recorded by the companies, so that the overall analysis of the popularity of these products can be done. The user ratings from 0-5 are logged so that the overall preference for the product can be evaluated. This data set can be evaluated using Apache Mahout in Eclipse IDE.

To integrate Java Code with Apache Mahout Libraries on Eclipse IDE, there are specific JAR files that are required to be added from Simple Logging Facade for Java (SLF4J).

The following is the Java Code module, with methods that can be executed using Eclipse IDE with the JAR files of Mahout to implement the recommender algorithm:

```
DataModel dm = new FileDataModel(new File("inputdata"));
UserSimilarity us = new PearsonCorrelationSimilarity(dm);
UserNeighborhood un = new ThresholdUserNeighborhood(Threshold
Value), us, dm);
UserBasedRecommender r=new GenericUserBasedRecommender(dm,
un, us);
List<RecommendedItem> rs=recommender.recommend(UserID,
Recommendations);
for (RecommendedItem rc : rs) {
System.out.println(rc);
}
```

Apache Mahout and R&D

Apache problems can be solved effectively using Apache Mahout with customised algorithms for multiple applications including malware predictive analytics, user sentiment mining, rainfall predictions, network forensics and network routing with deep analytics. Nowadays, the integration of deep learning approaches can be embedded in the existing algorithms so that a higher degree of accuracy and optimisation can be achieved in the results. 

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